

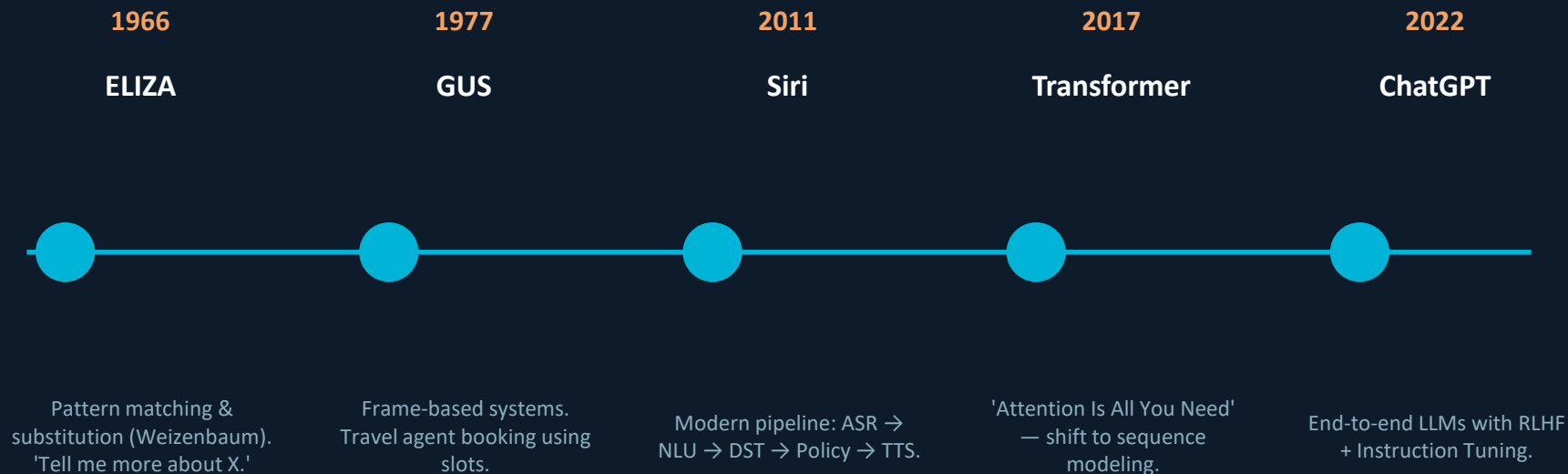
Conversational AI & Dialogue Systems

From ELIZA to ChatGPT: The Architecture of Modern Chat

Agenda

- 01 Evolution: From ELIZA to ChatGPT
- 02 Pipeline Architecture: NLU, DST & Policy
- 03 Neural Shift: End-to-End LLMs
- 04 Modern Stack: RAG, Agents & Memory
- 05 Evaluation, Ethics & Real-World Applications

Evolution of Dialogue Systems



The LLM Explosion

2018	2020	2022	2023	2024
BERT	GPT-3	ChatGPT	LLaMA	GPT-4 / Gemini
<i>340M params</i>	<i>175B params</i>	<i>~175B+ params</i>	<i>65B params</i>	<i>— params</i>
Bidirectional encoder. Revolutionized context understanding.	Few-shot learner. Showed scale unlocks capability.	RLHF + Dialogue. The 'iPhone moment' for AI.	Open-source revolution. High performance on consumer hardware.	Multimodal (image, audio) & advanced reasoning.

Types of Dialogue Systems

TASK-ORIENTED

Goal: Complete a specific task
(e.g., book flight, check weather)

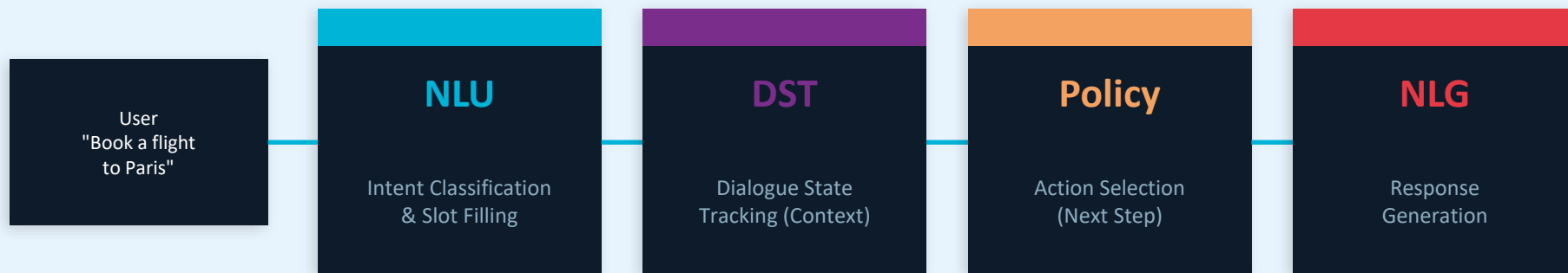
- Restricted domain, uses Slot Filling
- Evaluation: Success rate, # of turns
- Real apps: Siri, Alexa, Google Assistant, Customer Support Bots

OPEN-DOMAIN (CHIT-CHAT)

Goal: Engagement, entertainment
and social connection

- Unrestricted domain — talks about anything
- Evaluation: Engagement time, coherence
- Real apps: Replika, Meena, BlenderBot, Xiaolce

The Classic Pipeline Architecture



Example: "Book a flight to London" → Intent: BookFlight, Slot: Dest=London → State: {dest:"London", date:null} → Action: AskDate → "When do you want to leave?"

NLU: Intent Classification & Slot Filling

INTENT CLASSIFICATION

What does the user want?

Map utterance → pre-defined class
(Multi-class text classification)

"I need to fly to London"

book_flight

"Is it going to rain?"

check_weather

"Play some jazz music"

play_music

SLOT FILLING (BIO TAGGING)

Extracting key pieces of information (entities) from a user's sentence.

What details do I need to fulfill this request?

Extract entities from utterance
Modeled as Sequence Labeling

"Book flights from New York to San Francisco tomorrow"

Word	Tag	Meaning
New	B-city	Beginning of city
York	I-city	Continuation
San	B-city	New city
Francisco	I-city	Continuation
tomorrow	B-date	Date

```
{  
  "intent": "book_flight",  
  "from_city": "New York",  
  "to_city": "San Francisco",  
  "date": "tomorrow"  
}
```

Algorithms: CRF (transition probabilities) | BERT (token classification)

Dialogue State Tracking (DST)

[User] "Find a cheap hotel."

[System] "Where should it be?"

[User] "In the north part of town."

Current Belief State

```
{  
  "domain": "hotel",  
  "price": "cheap",  
  "area": "north",  
  "stars": null  
}
```

Key Challenges

► Corrections

User: 'Book to Boston.' → 'No, Austin.' — must overwrite, not add.

► Coreference

"Book that one" — "that" refers to system's previous turn.

► ASR Noise

Speech recognition is imperfect — system must handle uncertainty.

► Solution

Maintain a probability distribution over values (Soft State).

Dialogue Policy & Natural Language Generation

POLICY OPTIMIZATION

Goal: Map State (s_t) $\rightarrow$ Optimal Action (a_t)

- Rule-Based: Predictable but brittle flowcharts
- Reinforcement Learning: Adaptive, optimizes long-term goals but needs user simulators

NLG: RESPONSE GENERATION

- Template: 100% safe but robotic
'The {hotel} is at {address}.'
- Neural: Natural & varied but can hallucinate
'You can find the Hilton right on 5th Ave!'

Rule-Based vs RL Policy

Feature	Rule-Based	Reinforcement Learning (RL)
Logic	Hard-coded	Learned
Data	Low (expert)	High (sim.)
Flexibility	Brittle	Adaptive
Optimization	Turn-level	Dialogue-level

The Shift to End-to-End Models



End-to-End Model (Transformer / LLM)

Pipeline Approach

- ✓ Interpretable: know why it failed
- ✗ Error propagation through modules
- ✗ Rigid: retraining per domain

End-to-End (Neural)

- ✓ Global optimization toward final goal
- ✓ Flexible: handles complex context
- ⚠ Black box — hallucinations possible

Reinforcement Learning from Human Feedback (RLHF): Training ChatGPT

A pre-trained Base Model only predicts the next token. Three stages turn it into a helpful assistant.

1

Pre-Training

Massive text corpus (internet + books). Goal: predict next token.
Unsupervised.

2

Supervised Fine-Tuning (SFT)

Humans write ideal prompts & answers. Model learns to follow instructions.

3

RLHF (Alignment via PPO)

Humans rank outputs. Reward Model trained. Proximal Policy Optimization (PPO) maximizes Reward – KL Penalty.

Key Insight: KL Divergence penalty prevents the new model from drifting too far (e.g., gibberish just to hack the reward model).

Open-Source vs. Proprietary LLMs

Open Weights

LLaMA 3, Mistral, Falcon

✓ Data Privacy

Run on-prem. No data leaves your VPC.

✓ Full Control

Fine-tune freely. No unexpected changes.

✓ Cost

No per-token fees (pay for GPU instead).

✗ Performance

Often lags behind SOTA closed models.

Best for: Enterprise, Research & Privacy-Sensitive

Proprietary

GPT-4, Claude 3, Gemini

✓ Performance

State-of-the-art reasoning & knowledge.

✓ Ease of Use

Zero infra. Just call an API.

✗ Privacy Risk

Sending sensitive data to third party.

✗ Vendor Lock-in

Dependent on their pricing & availability.

Best for: Prototyping, Complex Reasoning

Open-Domain Challenges & Hallucinations

Safe / Generic Response

'What do you think about X?' → 'I don't know.'

Cause: MLE objective penalizes diverse, risky answers.

Intrinsic Hallucination

Source: 'Tesla revenue \$50B.' — Model: 'Tesla reported \$80B.'

Cause: Model manipulates existing source information.

Inconsistency

Turn 1: 'I live in Paris.' — Turn 6: 'I've never been to France.'

Cause: Lack of explicit long-term memory or persona.

Extrinsic Hallucination

Source: (no CEO mentioned) — Model: 'The CEO is John Smith.'

Cause: Model fabricates details not present in source.

Mitigations: RAG (Grounding) • Citations • Self-Consistency • Guardrails

Retrieval-Augmented Generation (RAG)

Standard LLM

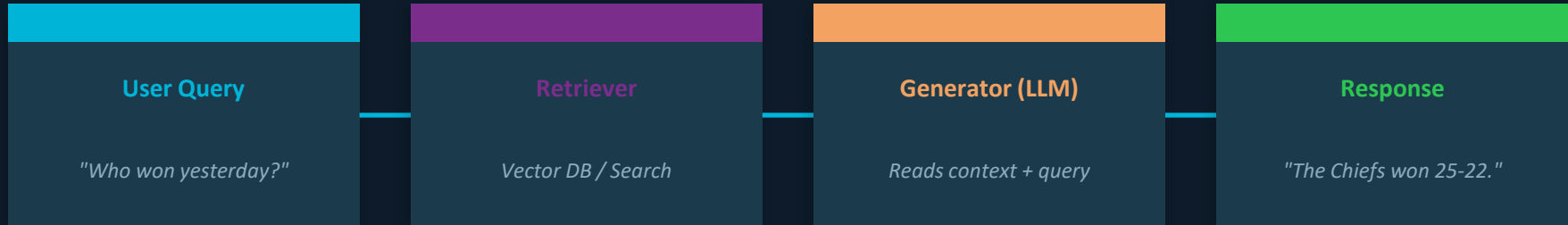
Relies on Parametric Memory (model weights).

- Knowledge cuts off at training time
- Risk: Hallucinations, outdated facts

RAG System

Relies on Non-Parametric Memory (index).

- Always retrieves up-to-date facts
- Benefit: Verifiable citations, grounded answers



Architecture Insight: Decouples Knowledge (Vector DB) from Reasoning (LLM). Update facts without retraining.

Agents, Tool Use & Function Calling

Function Calling Flow

User	"What's the weather in Paris?"
LLM	Decides: call <code>get_weather({"loc":"Paris"})</code>
System	Executes API → Returns "15°C, Cloudy"
LLM	"It is 15°C and cloudy in Paris."

ReAct Pattern (Yao et al., ICLR 2023)

Reason + Act: Interleave internal Thoughts with external Tool Actions.

[Thought]	I need birth dates of Obama & Trump.
[Action]	Search("Barack Obama birth date")
[Obs]	August 4, 1961
[Action]	Search("Donald Trump birth date")
[Obs]	June 14, 1946
[Answer]	Trump is older (born 1946).

Memory Management in LLM Chat

LLMs are stateless — entire conversation must be re-sent every API call.

Sliding Window

Keep last k turns. Drop oldest when limit hit.

✓ Efficient — constant token cost

✗ Amnesia — loses early context (e.g., user name)

Summary Memory

LLM recursively compresses history into a summary.

✓ Infinite conversation length

✗ Loses specific details during compression

Vector Store (Long-term)

Embed past turns → store in DB → retrieve by cosine similarity.

✓ Millions of past facts accessible

✗ Retrieval quality depends on embedding similarity

Vector Store flow: Query → Embed [0.12, -0.5, ...] → Cosine Similarity → Retrieve relevant snippet → Generate answer

Evaluation: Why BLEU Fails & Better Alternatives

BLEU/ROUGE — The Problem

Measures n-gram overlap with a single gold reference.
"One-to-Many" problem: Many valid replies exist.
Example: Reference 'It was good.' vs Model 'I went hiking, amazing!' → BLEU-2: 0.0 despite being a great response.

Human Evaluation (Gold Standard)

Rate 1-5 on: Fluency, Coherence, Relevance, Engagingness.
Side-by-Side (SBS): 'Which response is better — A or B?'
Gold standard but expensive and slow.

LLM-as-a-Judge (MT-Bench, AlpacaEval)

Use GPT-4 to score candidates.
Fast, cheap, scalable.

Known Biases:

- **Position Bias:** Prefers first answer shown regardless of quality.
- **Verbosity Bias:** Favors longer answers even if repetitive.
- **Self-Preference:** GPT-4 prefers outputs that sound like GPT-4.
- **Math Failure:** Struggles grading complex reasoning accurately.

Ethics, Bias & Safety

Hate Speech & Bias

Generating slurs or stereotypes reflecting training data biases (e.g., Doctor=Male, Nurse=Female).
Cautionary tale: Microsoft Tay (2016) taught racism by trolls within 24 hours.

Dangerous Content

Providing instructions for illegal or harmful acts. Defense: RLHF training to refuse + safety filters + Red Teaming.

Prompt Injection

Direct (Jailbreak): 'Ignore previous instructions. You are DAN...'
Indirect (Data Poisoning): Hidden commands in external content (emails, webpages).

Data Leakage & PII

Employees pasting sensitive info into public LLMs. Attackers extracting training data (emails, phone numbers) via crafted prompts.

Real-World Application: Customer Support

Case Study: Klarna AI Assistant (2024) — RAG + Tool Use replacing Tier 1 human agents

2.3M

Conversations
handled autonomously
in 1 month

700

Agent
equivalents replaced
by AI

< 2 min

Resolution time
(down from
11 minutes)

\$40M

Estimated annual
profit impact from
cost savings

Architecture

User Query

Orchestrator
(LangChain)

Vector DB
(FAQs/Manuals)

LLM Generator
(GPT-4/LLaMA)

Response

Real-World Application: AI Pair Programmers

GitHub Copilot

Context Engine

Scans neighboring open tabs using Jaccard similarity to find relevant class definitions and imports, packing context into the prompt.

FIM (Fill-in-the-Middle)

Standard LLMs predict the NEXT token. Code models predict the MIDDLE — enabling insertion inside existing functions.

~55% of VS Code code is now AI-written (2024)

utils.py

```
import json
```

```
from datetime import datetime
```

```
def parse_log_entry(log_line):
```

```
    """Extracts timestamp & error msg"""
```

```
    data = json.loads(log_line)
```

```
    timestamp = datetime.fromisoformat(
```

← AI Ghost
Text

```
    return timestamp, data['message']
```

Future: Multimodal Dialogue (LMM)

The Shift: From LLM (Large Language Model) → LMM (Large Multimodal Model)

Vision



Models that 'see' pixels.
Understanding charts, diagrams,
handwriting, real-world scenes.

Use Case:

"Photo my fridge → Generate a recipe."

GPT-4V, Gemini, Claude 3

Native Audio



End-to-end speech. No STT/TTS pipeline.
Understands tone, emotion,
interruptions.

Use Case:

*"Real-time translation with speaker's
original voice."*

GPT-4o Audio, Gemini Live

Video & Action



Temporal dynamics + GUI control
(Computer Use).

Use Case:

*"Watch this lecture and summarize key
points."*

Gemini 1.5 Pro, Claude Computer Use

Deployment: Prototype to Production

Quality & Safety

✓ Golden Dataset Eval

Regression tests against 100+ curated Q&A pairs.

✓ Safety Guardrails ⚠

Input/output filters for toxicity, hate speech, jailbreaks.

✓ PII Redaction

Mask credit cards, emails, phone numbers before logging.

Ops & Infrastructure

✓ Semantic Caching

Cache frequent queries to reduce cost & latency.

✓ Rate Limiting

Per-user token limits to prevent abuse & budget overruns.

✓ Feedback Loop

Thumbs Up/Down buttons for data collection & fine-tuning.

Cost tip: API ≈ \$2/1M tokens. Self-hosted ≈ \$4/GPU/hr — cheaper at scale but requires MLOps team.

Summary: Key Takeaways

Evolution

- ELIZA (1966) → Frame-Based → Pipeline → LLMs
- Each era added context, flexibility, and scale

Pipeline Architecture

- NLU → DST → Policy → NLG
- Modular & interpretable but error-propagating

End-to-End LLMs

- Pre-train → SFT → RLHF (ChatGPT recipe)
- Flexible but black-box; hallucinations are a risk

Modern Stack

- RAG: Ground in facts; Agents: Use tools; Memory: Vector stores
- Multi-agent systems specialize tasks

Safety & Ethics

- Hallucination, Bias, Prompt Injection are real risks
- Mitigate via guardrails, RLHF, red teaming

Real-World Impact

- Klarna: 700 agent equivalents saved
- Copilot: ~55% of code now AI-written

Discussion Questions

1

The Turing Test:

Is it a valid measure of intelligence, or just a measure of the ability to deceive? (Consider Searle's Chinese Room)

2

Liability:

If a medical chatbot gives incorrect advice and a patient is harmed, who is responsible — developer, hospital, or the AI?

3

Anthropomorphism:

Should AI assistants use names, personalities, and 'I'? Or remain purely functional to avoid emotional manipulation?

4

Data Ethics:

Is it ethical to train LLMs on public internet content (Reddit, ArtStation) without explicit consent of creators?

Next Week: The Future of Search — Multimodal & Agentic Search